

Sayantan Bhattacharya

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EDUCATION

Techno Main Salt Lake

B.Tech in Computer Science and Engineering (AI and Machine Learning)

Kolkata, India

GPA: 3.06 / 4.0

Expected June 2027

- Relevant coursework: Machine Learning, Deep Learning, NLP, Neural Networks, Data Structures and Algorithms, Probability and Statistics, Linear Algebra, Database Management Systems, Web Development, Computer Networks, Operating Systems

EXPERIENCE

Independent Technical Contractor

2020 — 2025

Content Systems and Automation — Remote

- Automated a multi-stage post-production pipeline using scripted DaVinci Resolve workflows, reducing per-project processing time by 40% across 50+ client deliverables — applied systematic optimization and iterative testing to identify and eliminate bottlenecks in a complex, multi-input system.
- Managed end-to-end delivery of 30+ long-form projects across international clients by building a structured scoping and revision framework, achieving zero late deliveries over 5 years.
- Designed reusable modular templates for color grading, motion, and audio processing by abstracting recurring patterns into parameterized workflows, scaling output 3x without proportional time increase.

PROJECTS

PassME — Full-Stack Password Manager

2025

Python, Flask, SQLite, HTML, CSS, JavaScript

CS50x Final Project

- Achieved zero critical vulnerabilities across all OWASP Top 10 attack vectors by implementing bcrypt hashing, Fernet encryption, and parameterized queries throughout the authentication and storage pipeline.
- Reduced credential retrieval time to under 100ms by designing an indexed SQLite schema with optimized CRUD operations, supporting password generation, copy-to-clipboard, and session management.
- Delivered a fully functional, security-compliant password manager as a solo developer, verified through manual penetration testing of all input surfaces.

Demo: youtube.com/watch?v=X2SUDquMxqc

Semantic Search RAG Chatbot — Two-Stage Retrieval Experiment

2026

Python, Flask, ChromaDB, LangChain, Groq (LLaMA-3.3 70B), Scrapy, HuggingFace Transformers

Personal Project

- Improved retrieval precision over single-stage cosine similarity search by designing a two-stage pipeline — ChromaDB vector search over 50 candidates followed by CrossEncoder reranking (ms-marco-MiniLM-L-6-v2) — demonstrating measurably more relevant rankings on ambiguous natural language queries.
- Reduced redundant re-indexing to near zero by building a Scrapy and Playwright crawler with SHA-256 content hashing, ensuring only changed pages trigger re-embedding across large document corpora.
- Enabled coherent multi-turn interaction via a conversation memory module preserving the last 3 Q&A pairs, generating cited answers using LLaMA-3.3 70B via Groq API across PDF, URL, and crawled web sources.

GitHub: github.com/PopcornKitsilano/rag-chatbot

Independent Study — Retrieval and Ranking Systems

2025 — 2026

Python, sentence-transformers, rank-bm25

Self-directed Research

- Reproduced BM25 vs dense retrieval comparison from the BEIR benchmark using rank-bm25 and sentence-transformers, confirming dense retrieval outperforms sparse on semantic similarity but underperforms on keyword-exact queries — findings directly informed the RAG chatbot architecture.
- Evaluated retrieval performance across 3 query types by designing a controlled experiment with consistent corpora, measuring nDCG@10 as the primary ranking metric.

Razer Crimson Royale — Custom Hardware Lighting Engine

2026

Python, Razer Chroma REST API

Personal Project

- Built a real-time per-key animation engine delivering 20 effects at 7fps by implementing HSV color math, directional transforms, and a session-based REST protocol — demonstrating applied systems programming outside coursework.
- Eliminated Razer Synapse dependency by reverse-engineering the local Chroma SDK REST API, implementing BGR encoding and a terminal-driven playlist system with full runtime control over effect order, speed, and direction.

GitHub: github.com/PopcornKitsilano/razer-crimson-royale

TECHNICAL SKILLS

Languages: Python, JavaScript, C, C++, Java, SQL, Bash

AI and ML: Machine Learning, Deep Learning, NLP, Computer Vision, Neural Networks, LLM frameworks, LangChain, Prompt Engineering, Model development and deployment, AWS SageMaker, AWS Bedrock, Conversational AI, Generative AI, Supervised and Unsupervised Learning, Feature Engineering

ML Libraries: NumPy, pandas, scikit-learn, Matplotlib, TensorFlow, PyTorch, Hugging Face Transformers

Web and Frameworks: Flask, Django, React, Node.js, HTML5, CSS3, Bootstrap, REST APIs

Databases: SQLite, SQL (schema design, indexing, joins, CRUD operations)

Cybersecurity: Penetration testing, OWASP Top 10, vulnerability assessment, SIEM, log analysis

Security Tools: BurpSuite, Metasploit, Wireshark, Hydra, CAPA, Splunk, Graylog

Developer Tools: Git, GitHub, Linux (Arch, Kali), DaVinci Resolve, Arduino, Agile, VS Code

Concepts: OOP, Data Structures and Algorithms, Software Design, Secure Coding

CERTIFICATIONS

CS50X — Introduction to Computer Science Harvard University via edX View Certificate Verified PDF	Aug 2025
Getting Started with Artificial Intelligence IBM via Credly View Certificate	Nov 2024
PreSecurity Certificate TryHackMe — 100+ rooms; real-world attack simulation, privilege escalation View Certificate	Jul 2025

TEST SCORES

- SAT (Scholastic Assessment Test): 1540 / 1600 | College Board | Dec 2021